# Accessible by Construction: Human-Centered Guardrails for Generative User Interfaces

NIHARIKA PRAMOD PUJARI, Independent Researcher, USA

Advances in artificial intelligence (AI) are rapidly reshaping design and development practices. Generative user interfaces (genUI) can now be produced partially or fully using modern AI systems driven by prompts, specifications, or examples. While these systems offer opportunities to create personalized and adaptive interfaces, they also introduce new risks for human-centered design, particularly around accessibility, ownership, user trust, and consistency. Generated interfaces may appear visually correct while silently failing to provide stable semantics, predictable interactions, or inclusive experiences.

In this paper, I argue that making generative user interfaces truly human-centered requires Human–Computer Interaction (HCI) and design practice to move from authoring interfaces toward defining constraints and review practices that guide how interfaces are generated. Using accessibility as a lens, I describe common interaction and accessibility failure modes in generative user interfaces, propose human-centered guardrails across generation and review stages, and identify open questions for HCI practitioners designing with generative interface systems.

Additional Key Words and Phrases: Generative user interfaces; Accessibility; Human–Computer Interaction; Human-centered AI; Design guardrails

## 1 Introduction: From Authored Interfaces to Generated Ones

Advances in generative AI have significantly transformed how user interfaces are created. Designers and developers are increasingly experimenting with systems that generate layouts, components, copy, and interaction patterns from natural language prompts, design tokens, screenshots, or structured specifications. This shift reframes interface creation from an authored activity to a synthesized one, where outputs may vary across contexts and generations [1].

From a human-centered perspective, generative user interfaces present compelling opportunities. Interfaces can adapt to individual needs, preferences, or abilities at a level of granularity that was previously difficult to achieve. However, these benefits challenge long-standing HCI assumptions that interfaces are carefully designed, reviewed, and finalized before reaching users. Generative systems can produce interfaces that appear visually convincing while remaining fragile in terms of interaction behavior, accessibility, and consistency.

This paper uses accessibility as an analytical lens to examine the challenges introduced by generative user interfaces. Accessibility violations often expose gaps between visual presentation and interactive or semantic behavior, making them a useful entry point for understanding how human-centered interaction must evolve in generative contexts.

## 2 Why Accessibility Is Uniquely Vulnerable in Generative UI

Traditional accessibility practices assume relatively stable interfaces, where components have known roles, labels, focus behavior, and interaction patterns. In generative systems, these properties may be implicitly inferred, inconsistently applied, or omitted altogether.

Early experimentation with generative user interfaces has revealed several recurring accessibility and interaction failure modes:

Author's Contact Information: Niharika Pramod Pujari, Independent Researcher, Westborough, Massachusetts, USA, niharika13pjr@gmail.com.

- **Label and name instability:** Generated content may duplicate, remove, or disconnect accessible names and instructions.
- **Inconsistent interaction patterns:** The same prompt may yield different interaction models, increasing cognitive load for users who rely on consistency [3].
- **Interaction unpredictability:** Focus order, keyboard navigation, and state transitions may vary across generations, undermining learning and trust.
- **Visual correctness versus semantic correctness:** Interfaces may appear accessible while lacking correct roles, names, or interaction semantics.

Users who rely on assistive technologies depend on predictable structure, consistency, and reliable feedback. These issues represent not only technical defects but also human-centered breakdowns. Without intentional constraints, generative user interfaces risk amplifying exclusion rather than reducing it.

## 3 New Human-Centered Experiences Enabled by Generative UI

When guided by human-centered principles, generative user interfaces can enable new forms of interaction aligned with established HCI goals. Rather than producing a single fixed interface, generative systems can create multiple versions tailored to a user's context, preferences, or abilities. For example, an interface might adjust its structure, interaction behavior, or information density based on situational or individual needs.

This adaptability has the potential to improve inclusion, user control, and personalization. However, these benefits emerge only when generation is guided by explicit human-centered expectations. Without such guidance, generative systems may prioritize speed or visual appeal over accessibility, predictability, and trust. This tension highlights the need for new design practices that balance flexibility with responsibility.

## 4 Guardrails as a Core HCI Practice for Generative UI

To support human-centered generative user interfaces, I propose treating guardrails as a core HCI practice. Guardrails define explicit constraints and review mechanisms that shape generative outputs toward inclusive and predictable behavior.

### 4.1 Generation-Time Constraints

Before interface generation, systems can be guided through structured prompts or specifications that encode human-centered requirements. These may include restricting generation to known accessible patterns, requiring accessible names for interactive elements, or specifying expectations for keyboard navigation and focus management. This approach shifts accessibility left, making it an integral part of generation rather than an afterthought.

### 4.2 Post-Generation Checks

Automated checks can provide rapid feedback on generated interfaces by validating minimum semantic requirements, detecting interaction traps, or comparing semantic changes across generations to surface regressions. While such checks cannot replace human judgment, they help scale oversight in dynamic generative systems.

### 4.3 Human Review and Accountability

Human review remains essential for assessing context, clarity, and predictability. Human-centered interaction practice must develop lightweight review guidelines for generated interfaces and clearly define responsibility when AI systems contribute to interface design. Clear review expectations help maintain trust and transparency for both users and practitioners.

## 5 Implications for HCI and Design Practice

Generative user interfaces challenge several long-standing assumptions in HCI and design practice.

**From authoring interfaces to defining rules:** Rather than crafting final interface artifacts, designers and HCI practitioners increasingly define guardrails and review practices that shape how interfaces are generated [2]. This shift requires new skills, tools, and evaluation methods.

**From static evaluation to continuous assessment:** Because generative interfaces may evolve over time, evaluation must consider ranges of generated outcomes, their consistency, and their impact on user trust.

**From individual authorship to shared responsibility:** Responsibility for interface quality spans designers, developers, and the systems they configure. HCI practice must clearly articulate how this responsibility is shared and communicated when interfaces are generated by AI.

Accessibility provides a concrete foundation for defining what "human-centered" means in generative contexts, offering lessons applicable to broader generative UI design challenges.

## 6 Contribution to the Workshop

At the W66 workshop, I aim to contribute:

- A preliminary set of common accessibility and interaction failure modes in generative user interfaces.
- Practical guardrail patterns applicable to real-world design and development workflows.
- Open questions around evaluation, responsibility, and user trust in generated interfaces.

Through discussion with workshop participants, I hope to refine these ideas and explore how HCI practice can proactively shape generative UI systems toward inclusive, human-centered outcomes.

## 7 Future Work

Future work will empirically evaluate the proposed guardrails through analysis of generated interfaces and expert walkthroughs. We also plan to study how different generation constraints influence semantic stability, interaction predictability, and perceived trust across iterations.

## References

[1] Saleema Amershi, Daniel Weld, Mihaela Vorvoreanu, Adam Fourney, Besmira Nushi, Penny Collisson, Jina Suh, Shamsi T. Iqbal, Paul N. Bennett, Kori Inkpen, Jaime Teevan, Ryen Kikin-Gil, and Eric Horvitz. 2019. Guidelines for Human–AI Interaction. In *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems* (Glasgow, Scotland, UK) *(CHI '19)*. ACM, New York, NY, USA, Article 3, 13 pages. doi:10.1145/3290605.3300233

[2] Ana-Elena Guriță and Radu-Daniel Vatavu. 2025. Breaking Bad (Design): Challenging AI User Interface Accessibility Guardrails. In *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems* (Yokohama, Japan) *(CHI EA '25)*. ACM, New York, NY, USA, Article 624, 7 pages. doi:10.1145/3706599.3716220

[3] Qian Yang, Aaron Steinfeld, Carolyn P. Rosé, and John Zimmerman. 2020. Re-examining Whether, Why, and How Human–AI Interaction Is Uniquely Difficult to Design. In *Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems* (Honolulu, HI, USA) *(CHI '20)*. ACM, New York, NY, USA, 12 pages. doi:10.1145/3313831.3376301